Reading the Table: Planning with Plan Recognition in Azul

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Abstract

Azul is a tile-drafting board game in which every tile, every pattern line and every wall is face up: the state is perfectly observable, and the only hidden thing is what the opponent *intends*. We show that this makes Azul an unusually clean testbed for plan recognition, provided one picks the right unit of intention. We define a plan as a *commitment of pattern lines to colours* and value a plan by projecting it onto the wall, so that adjacency and the row, column and colour bonuses are credited to the combination of commitments rather than to individual tiles. The abstraction is small enough to search exhaustively in principle and by a width-12 beam in practice (which recovers the exhaustive optimum on 98.3% of sampled boards), and concrete enough to be recognised from an opponent's public board: we maintain a Boltzmann-rational posterior over the opponent's commitments, updated by differencing their board between our own turns, and read it out as "what the opponent is waiting for". Tile denial then falls out of evaluating the position from both sides rather than being a bolted-on special case. A controlled ablation over 5,700 games separates two design decisions that are usually conflated. Modelling the opponent at all is worth 19.5 points of win rate against our heuristic baseline ($50.4\% \rightarrow 69.9\%$, $p < 10^{-11}$); recognising *which* plan they are pursuing adds a further increment that is small but reliable ($55.9\% \pm 2.4$ over 1,600 paired head-to-head games, $p < 10^{-5}$). Intrinsically, the recogniser names the opponent's next (line, colour) commitment 1.8–2.5 times as often as chance, and — the right sanity check — collapses to chance against a random opponent, who has no plan to find. To our knowledge this is the first published application of plan recognition to a tile-placement board game.

Introduction

Azul (Kiesling 2017) is a two- to four-player tile-drafting game that won the Spiel des Jahres in 2018 and has since become one of the most widely played modern board games. Its scoring is *spatial* and *deferred*: a tile is worth what its neighbours on the wall make it worth, and it is placed on the wall several rounds after it was drafted. That combination is exactly what defeats one-tile-at-a-time evaluation, and it is the structural property Azul shares with Carcassonne, the closest mechanically related game with a published agent (Valdez Ameneyro, Galván, and Kuri Morales 2020).

The observation this paper turns on, however, is a different one. Azul is a game of perfect information about *state* and no information at all about *intention*. Every factory display, every pattern line and every wall is face up. There is no hand, no hidden role, no fog of war. And yet what an opponent will take next is not determined by the visible state alone: it depends on commitments they have already made, on how badly they need a particular colour to complete a row they started three turns ago, and on which of several adequate continuations they are actually pursuing. Human Azul players do not compute; they *read the table*. They look at a half-filled pattern line and conclude that their opponent needs two more black tiles, and then they take the black tiles.

This is a clean setting for plan recognition, and cleaner than most. The hypothesis space is small and interpretable, because Azul's rules force a player to declare intent in public: a pattern line accepts exactly one colour, so filling one commits a row to a colour visibly and irrevocably for the rest of the round. The evidence is public and unambiguous — we see what a human player sees. And, unusually, there is a direct way to *act* on the inference, because there is a single shared tile supply: taking the tiles a recognised plan is waiting for is a legal, well-defined and often cheap move.

We exploit this with an agent that plans from both sides of the table. Our contributions are:

1. A plan abstraction for Azul — pattern-line commitments, valued jointly by projecting them onto the wall — that is cheap enough to search every turn and concrete enough to be recognised from an opponent's public board.
2. A Boltzmann-rational plan recogniser over that abstraction, with an intrinsic evaluation showing that the posterior is measurably informative: next-commitment prediction runs at 1.8–2.5 times chance, and falls to chance against an opponent with no plan.
3. An agent that evaluates each move from both sides of the table, so that tile denial emerges from the evaluation rather than being a special case.
4. A controlled ablation over three agents that share all of their code except one setting each, separating the value of *modelling the opponent at all* from the value of *recognising which plan they are pursuing*, with the honest finding that the first is large and the second small but reliable.
5. To the best of our knowledge, the first published application of plan recognition to a tile-placement or drafting board game, and the first refereed AI study of Azul.

We should be clear about the framing from the outset. Our baseline is our own one-ply heuristic, not the state of the art; a considerably stronger Azul agent already exists (Rzepecki 2025). The paper is about what opponent modelling buys in a domain where intention is the only hidden variable, not about absolute playing strength, and every number below should be read that way.

Related Work

AI for Modern Board Games

Modern tabletop games form a distinct class of AI domain — long horizons, more than two players, score-based rather than win/loss outcomes, and rules that resist the abstractions that work for chess and Go — and frameworks such as TAG (Gaina et al. 2020) and its reinforcement-learning successor PyTAG (Balla et al. 2024) exist to make that class tractable to study (Hu et al. 2024). Monte Carlo tree search is the default answer, applied to Settlers of Catan (Szita, Chaslot, and Spronck 2010), Carcassonne (Valdez Ameneyro, Galván, and Kuri Morales 2020) and many others (Browne et al. 2012; Świechowski et al. 2023). The Carcassonne work carries a point we reuse directly: when value accrues from spatial adjacency and from completing multi-tile structures, greedy evaluation is systematically weak, because the worth of a placement is not a property of the placement. Blood Bowl (Justesen et al. 2019) is the template for arguing that a particular board game deserves attention as a challenge domain, and Hanabi (Bard et al. 2020) is the reference argument that reasoning about *other agents' intentions* is a challenge distinct from search. Hanabi is cooperative and Azul is competitive, and the contrast is instructive: in Hanabi you infer intention in order to help, in Azul in order to deny.

On Azul specifically, the literature is thin to the point of absence. A systematic search of arXiv, Crossref, OpenAlex and DBLP returns no refereed conference or journal paper on AI for Azul. What exists is two student theses and a handful of frameworks. Počatko (2021) compares minimax with alpha–beta pruning, MCTS, and a knowledge-based strategy with iterative deepening; Rzepecki (2025) builds a minimax agent over handcrafted heuristics and then an NNUE-style learned evaluation that substantially outperforms it, and is the strongest Azul agent we are aware of. Blom's framework (2019) underpins a long-running university agent competition, and Ceramic (Swynfel 2020) is the most serious open-source research environment. None of these agents models the opponent's intentions, and none of the work is refereed. We therefore state the novelty claim in the checkable form: to the best of our knowledge, no peer-reviewed study of AI agents for Azul has been published, and no plan-recognition work exists in tile-placement games at all.

Plan and Goal Recognition

Plan recognition was posed as a formal problem by Kautz and Allen (1986): recover an agent's plan from partial observations of its behaviour, given a library of possible plans. Charniak and Goldman (1993) gave it a probabilistic treatment, maintaining a posterior over plans rather than committing to a single consistent explanation, and later work handles structured, partially executed plan libraries efficiently (Geib and Goldman 2009); see Sukthankar et al. (2014) and Van-Horenbeke and Peer (2021) for the field's own accounts of the activity/plan/goal distinction. Our agent is library-based in this sense, and what it infers is a structured multi-row commitment rather than a single terminal goal, which places it in plan rather than goal recognition.

The strand we actually build on is recognition-as-planning (Ramírez and Geffner 2009), in which the observer plans *for* the observed agent and compares the cost of each candidate goal with and without the observations, and in particular its probabilistic form (Ramírez and Geffner 2010), which yields a Boltzmann-style likelihood $P(O \mid G) \propto \exp(-\beta \Delta \mathrm{cost})$. Because planner-in-the-loop recognition is expensive, a body of work replaces it with cheaper evidence: landmarks (Pereira, Oren, and Meneguzzi 2017, 2020) and domain-specific cost formulae (Masters and Sardina 2017, 2019). Masters and Sardina's result — that replacing general planning with a domain-specific valuation is both faster and about as accurate — is the closest methodological precedent for what we do. Our setting is *online* (Vered and Kaminka 2017), since the posterior is updated incrementally as observations arrive, and *keyhole* (Avrahami-Zilberbrand and Kaminka 2007), since the observed agent is neither cooperating with nor hiding from us. See Meneguzzi and Pereira (2021) for a survey of the strand.

Opponent Modelling in Games

Albrecht and Stone (2018) separate *policy reconstruction* (predict what they will do) from *plan recognition* (infer what they are trying to achieve) from type reasoning. Our paper needs this distinction to describe its own ablation: `planner-bestresp` is essentially a best response to a re-planned opponent, and `planner` adds plan recognition on top. Poker is the classical demonstration that exploiting a specific opponent beats playing a fixed strong strategy (Billings et al. 2002), and Bayes' Bluff (Southey et al. 2005) maintains a posterior over opponent strategies and responds to it — structurally the same move we make. Synnaeve and Bessière (2011) recognise an opponent's opening in StarCraft and use the belief to select a counter-strategy, which is the closest published analogue to our whole pipeline, in RTS rather than a board game. Nashed and Zilberstein (2022) ask when opponent modelling actually pays, which is precisely the question our ablation is designed to answer.

Positioning

Our likelihood is a specialisation of inverse planning (Baker, Saxe, and Tenenbaum 2009; Ziebart et al. 2008) and of probabilistic recognition-as-planning (Ramírez and Geffner 2010), not a new inference method, and our agreement term is a hand-built landmark signal (Pereira, Oren, and Meneguzzi 2017). What is new is the plan abstraction, the domain, and the measurement. The plan recognition literature is dominated by navigation, robotics, RTS and intrusion detection; the board game literature is dominated by search and learning with little intention inference. Recognition-as-planning has not been applied to tile-placement board

games, and the two existing Azul agents model the opponent not at all.

The Game of Azul

We summarise the rules our engine implements; they are the standard printed rules (Kiesling 2017).

Round structure. With n players the table holds $2n + 1$ factory displays, each filled with four tiles drawn from a bag of 100 (20 in each of five colours). On their turn a player takes *all* tiles of one colour either from a single factory — the remaining tiles of that factory slide into the shared centre — or from the centre. The first player to take from the centre in a round also takes the first-player marker, which costs them a floor penalty and makes them the starting player next round. The round ends when factories and centre are empty.

Pattern lines and the wall. Each player has five pattern lines of capacity $1, \ldots, 5$, a 5×5 wall and a floor line. A pattern line holds tiles of a single colour, and may only hold a colour that the corresponding wall row does not already contain. Excess tiles from a take fall onto the floor line; dumping the whole take onto the floor is always legal. At the end of a round, each *full* pattern line moves one tile to its fixed square on the wall and discards the rest; partial lines simply stay. Each wall row is the colour sequence of the row above shifted one step right, so colour c occupies column $(c + r) \bmod 5$ in row r and every colour appears once per row and once per column.

Scoring. A tile placed on the wall scores the length of its contiguous horizontal run plus the length of its contiguous vertical run if it has neighbours in both directions; otherwise the single run it extends; otherwise one point. The floor line costs $-1, -1, -2, -2, -2, -3, -3$ for its seven spaces and a score never drops below zero. The game ends after the round in which some player completes a horizontal wall row, and is then followed by bonuses of 2 per completed row, 7 per completed column and 10 per colour placed five times. Ties are broken by completed rows. One point the rulebook leaves open: if the first-player marker is never taken during a round — possible only when the centre stays empty — we leave it unclaimed and the starting player unchanged.

What Makes Azul Hard

Four properties of the game shape the design that follows.

Scoring is joint. The value of a commitment depends on the other commitments it will sit next to. A column bonus is worth seven points and is created by five separate placements across five different rounds; no single placement is responsible for it. Per-tile evaluation therefore systematically under-values combinations, exactly as in Carcassonne (Valdez Ameneyro, Galván, and Kuri Morales 2020).

Scoring is deferred. A pattern line is a multi-round investment that pays nothing until it is complete, and is a pure liability if the round ends while it is not. Filling three of five squares of the bottom line is not 60% of the value; it is zero value and a hostage.

The supply is contested. There is one shared pool of tiles and taking is exclusive. Denial is therefore a real move rather than an afterthought, and its cost — the points you forgo by taking tiles you cannot use well — has to be weighed against its benefit.

Intentions are legible. A partially filled pattern line is a public, persistent, and largely irreversible declaration of intent. This is what makes recognition possible at all; in the vocabulary of goal recognition design (Keren, Gal, and Karpas 2019), Azul's rules already make the environment unusually revealing. Games are short (about 5 rounds) and the branching factor is modest (a mean of 25 legal moves per turn at two players, median 19, rising to a mean of 46 at four players), which together justify a design that re-plans from scratch every turn rather than maintaining a search tree.

Plans, Projection and Recognition

Plans as Pattern-Line Commitments

Let $\mathcal{C}$ be the five colours and $R = \{0, \ldots, 4\}$ the wall rows. A **plan** is a partial map $\pi : R \rightharpoonup \mathcal{C}$, written as a tuple $(c_0, \ldots, c_4)$ with $c_r \in \mathcal{C} \cup \{\bot\}$; we read $\pi = (\bot, \bot, \text{BLUE}, \text{WHITE}, \text{BLACK})$ as "finish line 3 in blue, line 4 in white, line 5 in black".

This is the right unit of intention for Azul for three reasons. It is the only piece of intention that *persists* across turns: which factory a player draws from is forgotten immediately, but a committed pattern line stays committed until it is completed or the round ends. It is *public*, so it can be recognised without any privileged access. And it is *small*: with five rows and five colours the space contains at most $6^5 = 7776$ plans, and far fewer once the wall fills, since a row may not be committed to a colour it already holds. Relative to classical planning (Ghallab, Nau, and Traverso 2004) this is a domain-specific abstraction rather than an action sequence, and it is closer in spirit to an HTN task network (Nau et al. 2003) — a set of commitments to be achieved — than to a STRIPS plan.

Valuing a Plan by Projection

A plan is valued by **projecting** it onto the wall. Its tiles are laid in tiling order (row 0 first) and scored against the wall *as it grows*, so that each commitment is scored in the context created by the earlier ones and the plan is credited for the adjacency and the bonuses its parts produce together.

Concretely, we carry the wall as a 25-bit integer W, bit $5r + c$ set when square (r, c) is occupied, so projecting a commitment is a handful of shifts. Let $\mathrm{col}(r, c)$ be the wall column of colour c in row r, $\mathrm{adj}(W, r, c)$ the Azul placement score of a tile at $(r, \mathrm{col}(r, c))$ on wall W, and $\mathrm{Bon}(W)$ the row/column/colour bonus of W. Writing B for the player's board, S for the tile supply currently on the table, and $\mathrm{need}(B, r, c)$ for the number of further c tiles row r requires, the value of one commitment is

$$v(W, r, c) = \delta(r, c)\,\big[\mathrm{adj}(W, r, c) + \mathrm{Bon}(W \cup s) - \mathrm{Bon}(W)\big] \tag{1}$$

with s the bit for $(r, \mathrm{col}(r, c))$, and the plan value is

$$V(B, \pi, S) = \sum_{r\,:\,\pi(r) \neq \bot} v(W_r, r, \pi(r)), \tag{2}$$

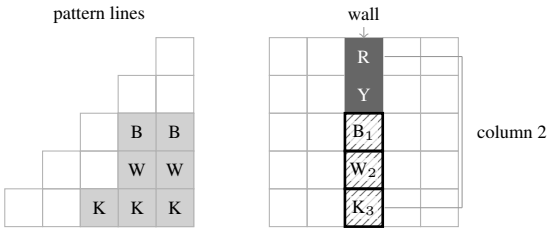


Figure 1: A plan projected onto the wall. Grey squares are already tiled; hatched squares are the plan $\pi = (\perp, \perp, \text{BLUE}, \text{WHITE}, \text{BLACK})$, projected in tiling order (subscripts). Scored jointly, BLUE extends the column to three (3 points, discounted 0.75^1), WHITE to four (4 points, 0.75^2) and BLACK closes it, scoring 5 plus the 7-point column bonus (0.75^2): $2.25 + 2.25 + 6.75 = 11.25$. Scored one at a time, the same three commitments are worth $2.25 + 0.5625 + 0.5625 = 3.375$. The column bonus belongs to no single commitment, which is exactly what a greedy evaluator cannot see.

where $W_0 = W_B$ is the player's current wall and $W_{r+1} = W_r \cup s_r$.

The discount $\delta(r, c)$ carries the two ideas that keep plans honest:

$$\delta(r,c) = \gamma^{\,\mathrm{need}(B,r,c)} \cdot \begin{cases} 1 & \text{if } S \text{ can supply it,} \\ \kappa & \text{otherwise.} \end{cases} \tag{3}$$

The **progress discount** $\gamma = 0.75$ makes a commitment worth γ^n of its face value when it still needs n tiles. Without it a plan is credited almost the same whether a row needs one more tile or five, so *making progress* means almost nothing and the agent drifts between half-finished commitments. It is the single most important parameter we tuned: adding it took the win rate against the heuristic from 61% to 74% on the tuning seeds, and it removed a diagnostic symptom worth reporting, because it tells you what was wrong. Before the discount, *widening* the candidate set made the agent worse (39% at 30 candidates rather than 10). A plan evaluator that degrades as it is given more moves to choose between is ranking moves worse than the cheap prefilter it is supposed to improve upon.

The **carryover** κ handles commitments the table cannot currently complete: if the supply of a colour, net of what earlier commitments in the same plan have already reserved, is short of what the row needs, the commitment keeps only $\kappa = 0.30$ of its value, and only 0.05 in the final round, when there is no later round left to cash it in. This is a crude treatment of the fact that the bag — unlike the table — is not observable; a principled treatment would determinise it (Cowling, Powley, and Whitehouse 2012).

Figure 1 shows why the joint scoring matters. The three commitments there are worth 11.25 together and 3.375 apart: a factor of 3.3, all of it invisible to an evaluator that scores one tile at a time.

Searching the Plan Space

Plans are found by a beam search over the rows in tiling order. The beam starts with the empty plan and, at row r, extends every surviving partial plan either by leaving r uncommitted or by committing it to each colour that is still available for that row and yields positive value; the extensions are ranked and the best $\beta_{\text{beam}} = 12$ are kept. Because rows are decided in the same order in which the projection scores them, the value of a partial plan is a genuine prefix of the value of every plan extending it, which is what makes a beam over this space well-behaved (Cazenave 2012).

The beam is a good approximation. Over 722 boards sampled from 361 states of `planner`-versus-`heuristic` games, the width-12 beam returned the exhaustive optimum on 98.3% of boards, and its value averaged 99.8% of the exhaustive optimum. We did not use an off-the-shelf classical planner (Helmert 2006), and the reason is worth stating to pre-empt the obvious objection: the plan space here is small, flat and highly structured, and it must be re-searched several dozen times per turn within a few milliseconds, which is the regime a domain-specific search is for.

Recognising the Opponent's Plan

Azul gives us perfect information about the opponent's state and none about their intention. We infer the latter with a posterior over the same plan space, generated for the opponent's own board.

Hypothesis space and prior. At each of our turns we run the beam search on the opponent's board and current supply and keep the top $|\mathcal{H}| = 12$ plans as hypotheses. The prior is the plans' own value, normalised: before seeing an opponent act, the best guess is that they are pursuing something good. When the hypothesis set is regenerated, the posterior mass of any hypothesis that survives is carried across, so evidence accumulates over the plans that remain plausible while the set tracks the changing board.

Observations. The engine shows an agent the state only at its own turns, so the opponent's moves are recovered by *differencing* their board between our turns: any pattern line that has grown yields a $\mathrm{Placement}(\ell, c, k)$ of k tiles of colour c on line ℓ, and tiles that appear only on the floor yield a floor placement. We see what a human player sees across the table, not a move list.

Update. For each observation o and hypothesis h,

$$\begin{aligned} \log P(o \mid h) = \beta\big[V(B', \pi_h, S) - V(B, \pi_h, S)\big] \\ + w \cdot \mathrm{agree}(\pi_h, o), \end{aligned} \tag{4}$$

where B and B' are the opponent's board before and after, and

$$\mathrm{agree}(\pi, o) = \begin{cases} +1 & \pi(\ell_o) = c_o, \\ -1 & \perp \neq \pi(\ell_o) \neq c_o, \\ 0 & \text{otherwise.} \end{cases} \tag{5}$$

The posterior is then renormalised in log space. Floor placements, and rows that π leaves uncommitted, contribute no agreement signal.

The first term is the Boltzmann-rational one: an agent pursuing π_h is exponentially more likely to take actions that advance it. It is a direct specialisation of the likelihood of Ramírez and Geffner (2010) and of inverse planning (Baker, Saxe, and Tenenbaum 2009; Ziebart et al. 2008), with their cost difference replaced by our plan-value difference and their inverse temperature kept as $\beta = 0.6$. It is worth noting what makes this term informative in Azul rather than merely positive: because a pattern line accepts one colour, a placement that puts red into a row π_h wanted for black does not merely fail to advance π_h, it *kills* that row's contribution outright, and the term goes sharply negative. Contradiction, not just confirmation, is observable.

The second term is a landmark-style agreement signal (Pereira, Oren, and Meneguzzi 2017, 2020): the row actually played into is a landmark for every hypothesis containing that commitment, and it separates hypotheses that remain merely consistent with the evidence, which the value difference alone can rank equally. We set $w = 1.2$.

Read-out. The posterior is summarised as *what the opponent is waiting for*. Marginalising over hypotheses gives a belief over commitments,

$$b(r, c) = \sum_{h \,:\, \pi_h(r)=c} P(h), \tag{6}$$

excluding rows whose wall square is already tiled, and from it an expected **demand**

$$d(c) = \sum_{r} b(r, c)\, \mathrm{need}(B_{\mathrm{opp}}, r, c). \tag{7}$$

Taking $\arg\max_{(r,c)} b(r, c)$ over rows the opponent has not yet filled gives a point prediction of their next commitment, which is the quantity we evaluate intrinsically later.

Acting on the Belief

Given a candidate move m that removes tiles from the table, let S be the supply before and S_m the supply after. The move's **denial** value against opponent j is the demand it puts out of reach:

$$D_j(m) = \sum_{c \in \mathcal{C}} \Big[\min\big(d_j(c), S(c)\big) - \min\big(d_j(c), S_m(c)\big) \Big]. \tag{8}$$

Only the shortfall counts: removing tiles an opponent does not want, or removing the fifth of a colour when they need two and eight remain, scores nothing.

This is the argument for why recognition adds something that re-planning cannot. Re-planning for the opponent tells you how well they *could* do from their current board; it is a statement about their opportunities. Recognition tells you what they are *already committed to*, and a commitment, unlike an opportunity, cannot be cheaply routed around: a pattern line half-filled with black is either finished in black or wasted.

The Decision Cycle

Each turn the agent performs four steps.

1. **Recognise.** Difference each opponent's board against the snapshot taken at our last turn, fold the resulting placements into that opponent's posterior via Eq. 4, and regenerate the hypothesis set from their current board.
2. **Filter.** Rank all legal moves with the cheap one-ply heuristic and keep the best $K = 10$. This is what keeps a decision at a few milliseconds: the plan search runs on ten candidates, not on all twenty-five to forty-six legal moves.
3. **Plan.** For each surviving move, apply it to a copy of our board, compute the supply it leaves behind, and re-plan from scratch, taking the value of the best plan available afterwards.
4. **Counter-plan.** Score the same position from the opponents' side and subtract it.

The move value is

$$\begin{aligned} U(m) = {} & V(B_m, \pi^*_m, S_m) + \mathrm{score}(B_m) + \mathrm{floor}(B_m) \\ & - \lambda_{\mathrm{opp}} \max_j V(B_j, \pi^*_j, S_m) \\ & + \lambda_{\mathrm{den}} \operatorname*{mean}_j D_j(m), \end{aligned} \tag{9}$$

with $\lambda_{\mathrm{opp}} = 0.5$ and $\lambda_{\mathrm{den}} = 0.7$, and the agent plays $\arg\max_m U(m)$. The threat term takes the *maximum* over opponents, which is a paranoid aggregation, while the denial term takes the *mean*, so that with three opponents the agent does not become a pure denial player; we return to this choice when we discuss the multi-player results.

The design point worth emphasising is that Eq. 9 contains no special case for denial. A move that starves a recognised plan scores well because the position is evaluated from both sides of the table, and the same expression that rewards building a column rewards preventing one.

Ablations by Construction

Two ablations differ from the full agent by exactly one setting each and share every other line of code.

- `planner-solo` drops the opponent model entirely: the last two terms of Eq. 9 vanish. It plans well for itself and ignores the table's other side.
- `planner-bestresp` keeps the threat term but drops recognition: it re-plans for each opponent, so it knows how well they *could* do, but it maintains no posterior and computes no denial term. In the vocabulary of Albrecht and Stone (2018) it is a best response to a re-planned opponent without plan recognition.

The comparison therefore isolates exactly two decisions: modelling the opponent at all (`planner-solo` versus `planner-bestresp`) and recognising which plan they are pursuing (`planner-bestresp` versus `planner`).

Cost. With n opponents, one decision runs $K(n+1)$ beam searches, each bounded by $5\beta_{\mathrm{beam}}|\mathcal{C}|$ commitment evaluations, and folds each new observation into $|\mathcal{H}|$ hypotheses at two plan valuations apiece. Everything is linear in the number of players.

Symbol	Meaning	Value
K	candidate moves after prefilter	10
β_{beam}	beam width over rows	12
γ	progress discount per missing tile	0.75
κ	carryover, unsupplied commitment	0.30
κ_{final}	carryover, final round	0.05
λ_{opp}	weight on opponent plan value	0.50
λ_{den}	weight on denial	0.70
β	inverse temperature of the likelihood	0.60
w	weight of the agreement term	1.20
$\lvert\mathcal{H}\rvert$	hypotheses in the posterior	12

Table 1: Tuned parameters, fixed for every experiment reported here, including the three- and four-player games.

Experimental Setup and Results

Setup

Engine and agents. We implemented the rules, the agent interface and the experiment harness ourselves in pure Python. Reusing an existing environment (Blom 2019; Swynfel 2020) was an option, but we wanted full control of cloning and determinisation and of the diagnostics the intrinsic recogniser evaluation needs — in particular the ability to ask an agent, at every one of its turns, what it believes about its opponent, and to score that belief against what the opponent then does.

Baselines. `random` picks uniformly among legal moves and serves only as a sanity floor. `heuristic` is a one-ply greedy agent over four hand-written features of the resulting board: the wall points its complete pattern lines would score (adjacency included), the fraction-filled progress of its incomplete lines, the floor penalty, and a smooth stand-in for the end-of-game bonuses that credits each row, column and colour set with its bonus scaled by the square of how full it is. It never looks at the opponent and never looks past the current move. It is also the prefilter inside the planning agents, which makes the comparison a fair one: the planner must beat the component it is built on top of.

Protocol. All matchups, game counts and seeds are fixed in a single script that takes no arguments. Seating is rotated between games so that each agent spends an equal share of games in each seat, cancelling the first-player advantage. A draw counts as half a win. Intervals are 95% normal-approximation intervals on the mean win share. Parameters were tuned on seeds 1000–1119 and evaluation begins at seed 50,000, in blocks disjoint across matchups, so no reported number comes from a tuning seed. The suite is 5,700 games and takes 42 s of wall time on a 24-core machine. Table 1 lists the tuned parameters.

We state plainly that all numbers below are relative to our own baseline and say nothing about absolute strength. A stronger Azul agent already exists (Rzepecki 2025), and the point of this evaluation is to attribute a difference between agents that share code, not to place any of them on an absolute scale.

Matchup (600 games)	Win rate	Score	Bonus
`planner`	$\mathbf{69.9}\% \pm 3.7$	53.5	11.0
vs `heuristic`	$30.1\% \pm 3.7$	44.5	5.3
`planner-bestresp`	$\mathbf{65.0}\% \pm 3.8$	55.5	11.7
vs `heuristic`	$35.0\% \pm 3.8$	49.0	6.2
`planner-solo`	$50.4\% \pm 4.0$	55.4	11.5
vs `heuristic`	$49.6\% \pm 4.0$	54.5	7.9

Table 2: Two-player results against the heuristic baseline. "Bonus" is the mean end-of-game row/column/colour bonus. Note that both ablations *score* more than the full agent and win less.

Two-Player Results

Both real agents beat `random` in 400 out of 400 games. That comparison establishes only a floor, and the reason is instructive about the game: random Azul scores about one point on average, because a uniform player dumps most of its tiles onto the floor line and the penalties cancel almost everything it manages to score. Azul punishes carelessness far more than it rewards cleverness.

The comparison that matters is against the heuristic. Over 600 games the full agent wins $69.9\% \pm 3.7$, with a mean score of 53.5 against 44.5 (Table 2). The two agents get there differently, and the column and row statistics show it: the heuristic completes about twice as many wall rows (1.67 versus 0.84) while the planner completes nearly five times as many columns (1.21 versus 0.26) and takes twice the end-of-game bonus (11.0 versus 5.3). This is the projection scoring doing exactly what it was designed to do. A greedy evaluator fills the short pattern lines because they complete soonest; a plan evaluator sees that three commitments spread across rows can line up into a column worth seven and is willing to wait for it.

What Each Component Is Worth

The ablation is the paper's main result, and it is lopsided; we state the lopsidedness up front rather than burying it.

Against the heuristic, `planner` wins 69.9%, `planner-bestresp` 65.0% and `planner-solo` 50.4% (Table 2, Figure 2). Head to head, where both agents see the same tile sequences, `planner` beats `planner-solo` $67.2\% \pm 3.8$ over 600 games and `planner-bestresp` $55.9\% \pm 2.4$ over 1,600 games. Two conclusions follow.

Modelling the opponent at all is where the advantage comes from. Removing the opponent model costs 19.5 points of win rate against the heuristic ($69.9\% \to 50.4\%$; two-sample $z = 7.05$, $p < 10^{-11}$), and head to head `planner-solo` loses $32.8\%/67.2\%$ ($z = 8.97$). Even the weaker form of the model — re-planning for the opponent with no recognition — is worth 14.6 points ($50.4\% \to 65.0\%$, $z = 5.17$, $p < 10^{-6}$). The instructive detail is in the score columns: `planner-solo` *scores more* than the full agent (55.4 versus 53.5) while winning far less. The points the full agent gives up in order to deny its opponent are not

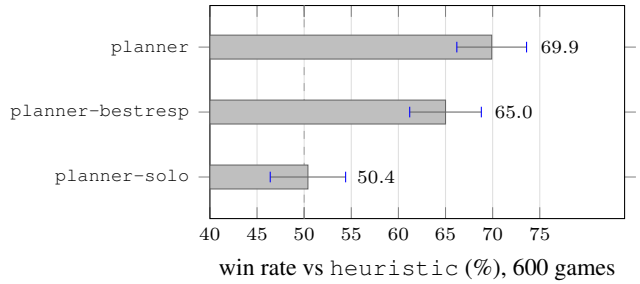


Figure 2: The ablation against the heuristic baseline, with 95% intervals. The step from `planner-solo` to `planner-bestresp` is the value of modelling the opponent at all; the step from `planner-bestresp` to `planner` is the value of recognising which plan they are pursuing. The dashed line is even.

wasted; they buy wins. An evaluation that looked only at mean score would have ranked these agents in the wrong order, which is a caution worth carrying to other score-based tabletop domains (Gaina et al. 2020).

Recognition adds a real but small increment. Against the heuristic the increment is $+4.9$ points of win rate ($65.0\% \rightarrow 69.9\%$), which on its own is not statistically resolved: a two-sample test on the two independent 600-game matchups gives $z = 1.82$, $p = 0.069$. The paired head-to-head comparison is what settles it. Over 1,600 games `planner` beats `planner-bestresp` 55.9%, with a 95% interval of $[53.5, 58.3]$ that excludes 50% comfortably ($z = 4.77$, $p < 10^{-5}$). At the 600 games the other matchups use, that interval still straddled 50%; the effect is reliable but it needed nearly three times the sample to separate from noise. We report this as the honest shape of the finding rather than quoting only the number that looks best.

Both agents already re-plan for the opponent, so both know how well the opponent could do. Recognition contributes exactly one thing on top: which colours the opponent is already committed to, and therefore cannot cheaply route around. A narrow addition is what the numbers show, and a likely reason it is narrow here is that the heuristic has no multi-row plan to recognise — it chooses one ply at a time. Recognising the intentions of an agent that does not have any is a known difficulty in this literature (Keren, Gal, and Karpas 2015), and it is notable that the increment is larger, not smaller, in the head-to-head column where the observed opponent does commit to plans.

Does the Recogniser Work?

Win rate is a blunt instrument for a component this specific, so we also evaluate the recogniser intrinsically, independently of whether it wins games. At each of its turns, before folding the newest placement into the posterior, the agent is asked to name the $(\text{line}, \text{colour})$ its opponent will commit to next; the answer is scored against what the opponent actually does. The chance baseline is a uniform guess over the placements still open to that opponent, computed per state, which is the fair comparison because the number of open placements shrinks as boards fill.

Opponent	Accuracy		Chance	Ratio
	line+col.	line	line+col.	
`random`	13.5	21.3	12.6	1.07
`heuristic`	34.8	49.3	19.3	1.81
`planner-solo`	31.2	36.3	14.2	**2.20**
`planner-bestresp`	29.6	34.2	14.7	**2.00**
3 players, 2 heur.	37.3	51.7	17.2	2.18
4 players, 3 heur.	36.9	51.1	14.8	2.50

Table 3: Intrinsic evaluation of the recogniser: percentage of turns on which the highest-posterior open commitment is the one the opponent makes next. Chance is a per-state uniform guess over the placements still open to that opponent. Between 5,740 and 34,392 predictions per row.

Table 3 gives the result across matchups. Against the heuristic the recogniser is correct on line *and* colour 34.8% of the time against a chance rate of 19.3%, and on line alone 49.3% — roughly 1.8 times chance over 12,177 predictions.

Two features of the table matter more than the headline number. First, the control: against `random` the accuracy collapses to 13.5% against a chance rate of 12.6%, a ratio of 1.07. This is the right outcome, and it is what tells us we are measuring recognition rather than a board-shape prior — there is no plan there to recognise, and the recogniser correctly finds nothing. Second, the recogniser does *better* relative to chance against opponents that actually plan: $2.20\times$ against `planner-solo` and $2.00\times$ against `planner-bestresp`, versus $1.81\times$ against the heuristic. This is direct evidence for the interpretation offered above, that the small win-rate increment against the heuristic reflects how little there is to read in a one-ply opponent rather than a failure of the inference.

Figure 3 tracks the posterior's entropy against the number of observations folded into it. It falls from 3.58 bits — close to the $\log_2 12 = 3.58$ of a uniform posterior over twelve hypotheses — to about 2.8 bits after a dozen observations. It is not monotone, because the hypothesis set is regenerated each turn from a board and a supply that keep changing, so evidence sharpens the belief while the world keeps widening it.

We want to be careful about what this figure does and does not show. The entropy falls at very nearly the same rate against a `random` opponent as against agents that plan, and against `random` we know from Table 3 that there is nothing to recognise. Entropy reduction is therefore *not* on its own evidence that the recogniser is picking up intention: some of it is simply the update concentrating mass on whichever hypotheses the accumulating board makes valuable. The accuracy comparison in Table 3, with its chance baseline and its random-opponent control, is the measurement that carries the claim. We report the entropy curves because they are the quantity an active recogniser would optimise (Shvo and McIlraith 2020), and because reporting the control curve as flat when it is not would be misleading.

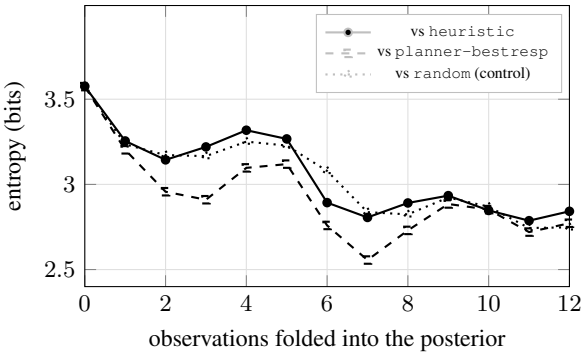


Figure 3: Posterior entropy against the number of observations. The uniform posterior over $|\mathcal{H}| = 12$ hypotheses is 3.58 bits. The belief sharpens with evidence but not monotonically, because the hypothesis set is regenerated each turn. The `random` control falls almost as fast, which is why the accuracy in Table 3, not this curve, is what demonstrates recognition.

Three and Four Players

Parameters were tuned at two players and left completely unchanged. The advantage survives with more seats but shrinks (Table 4): one planner against two heuristics wins $39.3\% \pm 5.5$ against a chance share of 33.3% ($z = 2.13$, $p = 0.033$), and against three heuristics $30.7\% \pm 5.2$ against a chance share of 25.0% ($z = 2.13$, $p = 0.033$). Both are positive and both are barely resolved at 300 games.

The negative result is worth reporting in full. A four-player field containing all three planning agents plus one heuristic does *not* order the agents as the two-player results do: the heuristic wins 35.7%, and the three planners finish in a statistical dead heat at the bottom (`planner-solo` 22.3%, `planner-bestresp` 21.3%, `planner` 20.7%). Three explanations are available and we cannot separate them with the data we have.

First, the parameters are untuned for this setting, and Eq. 9 in particular is a two-player expression pressed into multi-player service: the threat term takes a maximum, which is paranoid, while the denial term takes a mean, and neither aggregation was chosen with three opponents in mind. Second, and more fundamentally, the cost of denial in a multi-player game is paid by one player and the benefit is shared by all the others — the standard difficulty of multi-player search (Sturtevant 2003, 2008). Three agents that each spend points denying, in a field with one agent that spends nothing on it, is close to a worst case for that asymmetry, and it is consistent with the heuristic's much higher mean score (60.6 against 54.8–55.5). Third, 300 games cannot resolve gaps of one or two points between three near-identical agents. A leader-targeting formulation (Goodman, Perez-Liebana, and Lucas 2023) rather than an opponent-summed one is the obvious correction and we have not tried it.

Cost

Table 5 reports single-core timings measured on an Intel-class desktop core; the 42 s figure for the full suite is wall time across 24 worker processes. A full-agent decision costs a mean of 2.84 ms at two players (median 2.56, 95th percentile 6.6), against 0.14 ms for the heuristic. Roughly half of that is the price of also planning for the opponent: `planner-solo` decides in 1.33 ms. Recognition itself is cheap — 2.46 ms for `planner-bestresp` versus 2.84 ms with recognition, so about 0.4 ms, or 13% of a decision — because the posterior is twelve hypotheses over a plan valuation that is a few shifts on a bitboard.

The prefilter is what keeps the cost in this range. Without it the plan search would run on all 25 legal moves at two players and all 46 at four, rather than on 10; the four-player decision cost of 7.4 ms is already dominated by planning three opponents rather than one. For comparison, an MCTS agent (Browne et al. 2012) at a comparable level would spend this budget on a few hundred playouts of a five-round game — possible, but it would not produce an inspectable statement about what the opponent is doing.

Agent	Win rate	Chance
3 players: 1 planner vs 2 heuristics		
`planner`	$\mathbf{39.3}\% \pm 5.5$	33.3%
`heuristic`	$30.3\% \pm 3.7$	
4 players: 1 planner vs 3 heuristics		
`planner`	$\mathbf{30.7}\% \pm 5.2$	25.0%
`heuristic`	$23.1\% \pm 2.7$	
4 players: the whole ablation		
`heuristic`	$\mathbf{35.7}\% \pm 5.4$	25.0%
`planner-solo`	$22.3\% \pm 4.7$	
`planner-bestresp`	$21.3\% \pm 4.6$	
`planner`	$20.7\% \pm 4.6$	

Table 4: Three- and four-player results (300 games each) with two-player parameters. Heuristic win rates are pooled over the identical heuristic seats. The bottom block is a negative result: with every agent paying the cost of denial and its benefit shared, the field inverts.

Agent (matchup)	Mean	Median	p95
`heuristic` (2p)	0.14	0.11	0.34
`planner-solo` (2p)	1.33	1.21	3.02
`planner-bestresp` (2p)	2.46	2.25	6.12
`planner` (2p)	2.84	2.56	6.61
`planner` (4p)	7.39	6.91	14.96

Table 5: Milliseconds per decision, single core, measured over 40 games per row against `heuristic` opponents. A two-player game is about 27 decisions per agent.

Threats to Validity

The baseline is our own, and the planning agents use it as their prefilter, so the comparison measures improvement over a component rather than standing. The parameters were tuned against that same heuristic and may be co-adapted to it; the head-to-head ablations, where both sides share the tuning, are the safer numbers. The recogniser is evaluated mostly against opponents that do not plan, which is pre-

cisely the regime in which recognition has least to offer, so the win-rate increment we report is plausibly a lower bound rather than an estimate. Multi-player parameters are untuned and the multi-player results should be read as an existence claim, not a measurement. Finally, no human play data enters anywhere: whether the recogniser's inferences resemble what human players infer is an open question that our evaluation cannot address.

Conclusion and Future Work

We set out to test a single idea: that in a game where the state is public and only intention is hidden, intention can be recognised and acted on if one picks the right unit for it. In Azul that unit is the pattern-line commitment. It is what the rules force a player to declare in public and then stick with, it persists long enough to be worth reasoning about, and the space of such commitments is small enough to search every turn and interpretable enough that the resulting belief can be read in plain language — "the opponent is two black tiles short of finishing line 5".

Around that abstraction we built a knowledge-light agent that trains on nothing, decides in under three milliseconds, and beats its own greedy baseline in 69.9% of 600 games by valuing combinations of commitments rather than tiles. More useful than the win rate is the attribution. A controlled ablation over 5,700 games shows that modelling the opponent at all is worth 19.5 points of win rate, while recognising *which* plan they pursue adds a further increment that is small but reliable — $55.9\% \pm 2.4$ over 1,600 paired games, an effect that took nearly three times the usual sample to separate from noise. Intrinsically the recogniser predicts the opponent's next commitment at 1.8–2.5 times chance and collapses to chance against an opponent with no plan, which is the control that makes the rest of the measurement meaningful.

We think the uncomfortable half of that answer is the more interesting half. Opponent modelling is often reported as a single capability that either helps or does not; splitting it into "model them" and "read them" shows the two halves differ by an order of magnitude in this domain, and that the smaller half grows against opponents who actually have plans. Reporting that recognition buys little against an opponent with nothing to recognise is not a disappointing result about the method; it is a statement about how much intention-reading is worth in Azul, and the sort of statement the opponent-modelling literature has asked for (Nashed and Zilberstein 2022).

The methodological point that may travel beyond this game is the one the plan abstraction rests on. Many games and many multi-agent settings have some mechanism that forces an agent to declare part of its intention publicly and bear a cost for abandoning it — a build order, a contract, a committed resource, a partially assembled structure. Where such a mechanism exists, it is a better hypothesis space for a recogniser than the space of goals or the space of policies, because the environment has already done the work of making the commitment observable and costly to reverse.

Future work, roughly in order of expected value:

Recognise a planning opponent. Our recogniser was measured mostly against agents with nothing to recognise, and it did visibly better relative to chance against the ones that plan. Self-play and a round-robin against agents that commit to multi-row plans is the experiment that would place the recognition increment properly, and we expect it to be larger than the number we report.

Fix the multi-player case. Re-tune at three and four players, and replace the max-threat/mean-denial aggregation of Eq. 9 with a paranoid, $\max^n$ or leader-targeting formulation (Sturtevant 2003; Goodman, Perez-Liebana, and Lucas 2023) that accounts for denial benefiting third parties. The four-player field in which every planner loses to the heuristic is, we suspect, diagnostic of exactly that asymmetry rather than of the abstraction.

Learn the plan valuation. The discounts and weights in Table 1 are hand-set. Replacing them with learned parameters, or replacing the projection score entirely with an NNUE-style evaluation (Rzepecki 2025) used as the plan scorer, would combine a learned evaluation with a symbolic hypothesis space and keep the belief interpretable — which is the property the AlphaZero recipe (Silver et al. 2018) gives up.

Stronger baselines. An ISMCTS (Cowling, Powley, and Whitehouse 2012) or determinised search baseline, and a comparison against the published NNUE agent if it can be obtained, would place the agent on an absolute scale rather than a relative one. Folding the posterior into the determinisation of a sampling search is also the natural way to combine the two approaches rather than choosing between them.

Active and adversarial recognition. Our agent recognises passively and then exploits. Choosing moves partly for what they reveal (Bisson et al. 2011; Shvo and McIlraith 2020) is well defined in Azul, since taking a tile an opponent may need forces a revealing response, and the converse — play that makes one's own commitments harder to read — becomes interesting as soon as both players recognise.

Human studies and release. Whether the recogniser's predictions agree with what human players report inferring is a question the game is unusually well suited to, and a legible belief over commitments is a natural basis for a teaching or explanation tool. We intend to release the engine, the agents and the fixed comparison suite so that the results are reproducible and the domain is reusable.

References

Albrecht, S. V.; and Stone, P. 2018. Autonomous Agents Modelling Other Agents: A Comprehensive Survey and Open Problems. *Artificial Intelligence*, 258: 66–95.

Avrahami-Zilberbrand, D.; and Kaminka, G. A. 2007. Incorporating Observer Biases in Keyhole Plan Recognition (Efficiently!). In *Proceedings of the 22nd AAAI Conference on Artificial Intelligence*, 944–949.

Baker, C. L.; Saxe, R.; and Tenenbaum, J. B. 2009. Action Understanding as Inverse Planning. *Cognition*, 113(3): 329–349.

Balla, M.; Long, G. E. M.; Goodman, J.; Gaina, R. D.; and Perez-Liebana, D. 2024. PyTAG: Tabletop Games for Multiagent Reinforcement Learning. *IEEE Transactions on Games*, 16(4): 993–1002.

Bard, N.; Foerster, J. N.; Chandar, S.; Burch, N.; Lanctot, M.; Song, H. F.; Parisotto, E.; et al. 2020. The Hanabi Challenge: A New Frontier for AI Research. *Artificial Intelligence*, 280: 103216.

Billings, D.; Davidson, A.; Schaeffer, J.; and Szafron, D. 2002. The Challenge of Poker. *Artificial Intelligence*, 134(1–2): 201–240.

Bisson, F.; Kabanza, F.; Benaskeur, A. R.; and Irandoust, H. 2011. Provoking Opponents to Facilitate the Recognition of Their Intentions. In *Proceedings of the 25th AAAI Conference on Artificial Intelligence*, 1762–1763.

Blom, M. 2019. AZUL: A Framework to Support Policy Learning for the Board Game Azul. https://github.com/michelleblom/AZUL. Used in COMP90054 AI Planning for Autonomy, University of Melbourne.

Browne, C. B.; Powley, E.; Whitehouse, D.; Lucas, S. M.; Cowling, P. I.; Rohlfshagen, P.; Tavener, S.; Perez, D.; Samothrakis, S.; and Colton, S. 2012. A Survey of Monte Carlo Tree Search Methods. *IEEE Transactions on Computational Intelligence and AI in Games*, 4(1): 1–43.

Cazenave, T. 2012. Monte Carlo Beam Search. *IEEE Transactions on Computational Intelligence and AI in Games*, 4(1): 68–72.

Charniak, E.; and Goldman, R. P. 1993. A Bayesian Model of Plan Recognition. *Artificial Intelligence*, 64(1): 53–79.

Cowling, P. I.; Powley, E. J.; and Whitehouse, D. 2012. Information Set Monte Carlo Tree Search. *IEEE Transactions on Computational Intelligence and AI in Games*, 4(2): 120–143.

Gaina, R. D.; Balla, M.; Dockhorn, A.; Montoliu, R.; and Perez-Liebana, D. 2020. Design and Implementation of TAG: A Tabletop Games Framework. *arXiv preprint arXiv:2009.12065*.

Geib, C. W.; and Goldman, R. P. 2009. A Probabilistic Plan Recognition Algorithm Based on Plan Tree Grammars. *Artificial Intelligence*, 173(11): 1101–1132.

Ghallab, M.; Nau, D.; and Traverso, P. 2004. *Automated Planning: Theory and Practice*. Morgan Kaufmann.

Goodman, J.; Perez-Liebana, D.; and Lucas, S. M. 2023. Following the Leader in Multiplayer Tabletop Games. In *Proceedings of the 18th International Conference on the Foundations of Digital Games (FDG)*.

Helmert, M. 2006. The Fast Downward Planning System. *Journal of Artificial Intelligence Research*, 26: 191–246.

Hu, C.; Zhao, Y.; Wang, Z.; Du, H.; and Liu, J. 2024. Games for Artificial Intelligence Research: A Review and Perspectives. *IEEE Transactions on Artificial Intelligence*, 5(12): 5949–5968.

Justesen, N.; Uth, L. M.; Jakobsen, C.; Moore, P. D.; Togelius, J.; and Risi, S. 2019. Blood Bowl: A New Board Game Challenge and Competition for AI. In *IEEE Conference on Games (CoG)*, 1–8.

Kautz, H. A.; and Allen, J. F. 1986. Generalized Plan Recognition. In *Proceedings of the 5th National Conference on Artificial Intelligence (AAAI)*, 32–37.

Keren, S.; Gal, A.; and Karpas, E. 2015. Goal Recognition Design for Non-Optimal Agents. In *Proceedings of the 29th AAAI Conference on Artificial Intelligence*, 3298–3304.

Keren, S.; Gal, A.; and Karpas, E. 2019. Goal Recognition Design in Deterministic Environments. *Journal of Artificial Intelligence Research*, 65: 209–269.

Kiesling, M. 2017. Azul. Board game, Plan B Games / Next Move Games.

Masters, P.; and Sardina, S. 2017. Cost-Based Goal Recognition for Path-Planning. In *Proceedings of the 16th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 750–758.

Masters, P.; and Sardina, S. 2019. Cost-Based Goal Recognition in Navigational Domains. *Journal of Artificial Intelligence Research*, 64: 197–242.

Meneguzzi, F.; and Pereira, R. F. 2021. A Survey on Goal Recognition as Planning. In *Proceedings of the 30th International Joint Conference on Artificial Intelligence (IJCAI)*, 4524–4532.

Nashed, S. B.; and Zilberstein, S. 2022. A Survey of Opponent Modeling in Adversarial Domains. *Journal of Artificial Intelligence Research*, 73: 277–327.

Nau, D. S.; Au, T.-C.; Ilghami, O.; Kuter, U.; Murdock, J. W.; Wu, D.; and Yaman, F. 2003. SHOP2: An HTN Planning System. *Journal of Artificial Intelligence Research*, 20: 379–404.

Pereira, R. F.; Oren, N.; and Meneguzzi, F. 2017. Landmark-Based Heuristics for Goal Recognition. In *Proceedings of the 31st AAAI Conference on Artificial Intelligence*, 3622–3628.

Pereira, R. F.; Oren, N.; and Meneguzzi, F. 2020. Landmark-Based Approaches for Goal Recognition as Planning. *Artificial Intelligence*, 279: 103217.

Počatko, M. 2021. *Artificial Intelligence for the Game of Azul*. Bachelor's thesis, Charles University, Faculty of Mathematics and Physics, Prague, Czech Republic.

Ramírez, M.; and Geffner, H. 2009. Plan Recognition as Planning. In *Proceedings of the 21st International Joint Conference on Artificial Intelligence (IJCAI)*, 1778–1783.

Ramírez, M.; and Geffner, H. 2010. Probabilistic Plan Recognition Using Off-the-Shelf Classical Planners. In *Proceedings of the 24th AAAI Conference on Artificial Intelligence*, 1121–1126.

Rzepecki, M. 2025. *Implementing Superhuman AI for Azul Board Game with a Variation of NNUE*. Master's thesis, University of Wrocław, Faculty of Mathematics and Computer Science.

Shvo, M.; and McIlraith, S. A. 2020. Active Goal Recognition. In *Proceedings of the 34th AAAI Conference on Artificial Intelligence*, 9957–9966.

Silver, D.; Hubert, T.; Schrittwieser, J.; Antonoglou, I.; Lai, M.; Guez, A.; et al. 2018. A General Reinforcement Learning Algorithm that Masters Chess, Shogi, and Go Through Self-Play. *Science*, 362(6419): 1140–1144.

Southey, F.; Bowling, M.; Larson, B.; Piccione, C.; Burch, N.; Billings, D.; and Rayner, C. 2005. Bayes' Bluff: Opponent Modelling in Poker. In *Proceedings of the 21st Conference on Uncertainty in Artificial Intelligence (UAI)*, 550–558.

Sturtevant, N. 2003. A Comparison of Algorithms for Multi-player Games. In *Computers and Games (CG 2002)*, volume 2883 of *Lecture Notes in Computer Science*, 108–122.

Sturtevant, N. 2008. An Analysis of UCT in Multi-player Games. *ICGA Journal*, 31(4): 195–208.

Sukthankar, G.; Geib, C.; Bui, H. H.; Pynadath, D. V.; and Goldman, R. P., eds. 2014. *Plan, Activity, and Intent Recognition: Theory and Practice*. Morgan Kaufmann.

Świechowski, M.; Godlewski, K.; Sawicki, B.; and Mańdziuk, J. 2023. Monte Carlo Tree Search: A Review of Recent Modifications and Applications. *Artificial Intelligence Review*, 56: 2497–2562.

Swynfel. 2020. Ceramic: An Environment to Play Variations of the Board Game Azul. https://github.com/Swynfel/ceramic.

Synnaeve, G.; and Bessière, P. 2011. A Bayesian Model for Plan Recognition in RTS Games Applied to StarCraft. In *Proceedings of the 7th AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE)*, 79–84.

Szita, I.; Chaslot, G.; and Spronck, P. 2010. Monte-Carlo Tree Search in Settlers of Catan. In *Advances in Computer Games (ACG 2009)*, volume 6048 of *Lecture Notes in Computer Science*, 21–32.

Valdez Ameneyro, F.; Galván, E.; and Kuri Morales, A. F. 2020. Playing Carcassonne with Monte Carlo Tree Search. In *IEEE Symposium Series on Computational Intelligence (SSCI)*, 2343–2350.

Van-Horenbeke, F. A.; and Peer, A. 2021. Activity, Plan, and Goal Recognition: A Review. *Frontiers in Robotics and AI*, 8: 643010.

Vered, M.; and Kaminka, G. A. 2017. Heuristic Online Goal Recognition in Continuous Domains. In *Proceedings of the 26th International Joint Conference on Artificial Intelligence (IJCAI)*, 4447–4454.

Ziebart, B. D.; Maas, A. L.; Bagnell, J. A.; and Dey, A. K. 2008. Maximum Entropy Inverse Reinforcement Learning. In *Proceedings of the 23rd AAAI Conference on Artificial Intelligence*, 1433–1438.